

1 Revision of basic thermodynamics – solutions

Problem 1.1 Ideal gas revision

Numerical answers in square brackets.

- (i) What volume does 1 kilomole of gas occupy at STP? 22.4 m^3 (general knowledge of ideal gas law).
- (ii) How many molecules does 1 mol of gas contain? Avogadro's number $= 6.022 \times 10^{23}$ (general knowledge).
- (iii) A 5 litre vessel contains 3 mol of gas at 200 K. What is the pressure?

$$pV = nRT \Rightarrow p = \frac{3 \times 8.31 \times 200}{5 \times 10^{-3}} = 9.97 \times 10^5 \text{ Pa}.$$

- (iv) 3 mol of an ideal monoatomic gas is at a temperature of 60°C . What is (noting a monotonic gas has 3 degrees of freedom excited at room temperature)

- (a) the internal energy, U ?

$$U = \frac{3}{2}nRT = \frac{3}{2}3 \times 8.31 \times (273 + 60) = 12.5 \text{ kJ}.$$

- (b) the heat capacity at constant volume,

$$C_V = \frac{3}{2}nR = 37.4 \text{ J K}^{-1}.$$

- (c) the heat capacity at constant pressure?

$$C_p = C_V + nR = 62.3 \text{ J K}^{-1}.$$

- (d) why is C_p greater than C_V ? This is because increasing the temperature at constant pressure increases the volume, which involves doing some work $\int p dV$.

Problem 1.2 Balloon

A balloon contains 200 kg of helium gas. At launch the pressure is 1030 mbar and the temperature is 17°C . Assuming that the pressure in the balloon is the same as the surrounding air:

- (i) What is the volume of the balloon?

$$n = \frac{200 \text{ kg}}{4 \times 10^{-3}} = 50 \times 10^3 \text{ mol} \Rightarrow V_1 = \frac{nRT}{p} = \frac{50 \times 10^3 \times 8.3 \times (273 + 17)}{1.03 \times 10^5} = 1.17 \times 10^3 \text{ m}^3.$$

- (ii) When the balloon has risen to its maximum height, where the pressure is 200 mbar and the temperature is only 5°C , what is the new volume?

$$\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2} \Rightarrow V_2 = \frac{p_1 T_2 V_1}{p_2 T_1} = \frac{1030(273 + 5)}{200(273 + 17)} 1.17 \times 10^3 = 5.78 \times 10^3 \text{ m}^3.$$

- (iii) Determine a relation between the difference of pressures inside, p_i , and outside, p_o , the balloon, the radius of the balloon, R and the surface energy (or surface tension) of the balloon, σ . What does the assumption about the equality of pressures inside and outside imply about the material the balloon is made of? To have identical pressures inside and outside the balloon, $\Delta p = p_{in} - p_{out} = 0$, the tension in the surface of the balloon must be negligible. To see this note work done in inflating the balloon by dV is $\Delta p dV = (p_i - p_o)dV$ and this must equal the increase of the surface energy, σdA , where σ is the tension in the balloon. Now

$$\begin{aligned} dV &= \frac{4\pi}{3} [(R + dR)^3 - R^3] = 4\pi R^2 dR \\ dA &= 4\pi [(R + dR)^2 - R^2] = 8\pi R dR . \\ \Rightarrow \Delta p \cdot 4\pi R^2 dR &= \sigma \cdot 8\pi R dR \\ \Rightarrow \Delta p &= \frac{2\sigma}{R} . \end{aligned}$$

Thus for $\Delta p = 0$, $\sigma = 0$ or $R \rightarrow \infty$.

Problem 1.3 Isothermal and adiabatic processes

0.5 m³ of gas at 4 bar is compressed until the pressure is 8 bar. Given that $\gamma = 1.4$, what is the new volume if

- (i) the process is isothermal? The ideal gas equation of state at constant T gives $p_1 V_1 = p_2 V_2$, therefore the volume after compression is

$$V_2 = \frac{p_1}{p_2} V_1 = 0.5 \times 0.5 = 0.25 \text{ m}^3 .$$

- (ii) the process is isentropic? Adiabatic transformation:

$$p_1 V_1^\gamma = p_2 V_2^\gamma \Rightarrow V_2 = V_1 \left(\frac{p_1}{p_2} \right)^{1/\gamma} = 0.5 \left(\frac{1}{2} \right)^{1/1.4} = 0.30 \text{ m}^3 .$$

- (iii) What does the value of γ imply if the gas is ideal? We know that for an ideal gas with m degrees of freedom,

$$\frac{7}{5} = \gamma = \frac{(m+2)(R/2)}{m(R/2)} \Rightarrow 7m = 5(m+2) \Rightarrow m = 5 ,$$

since we assume three dimensions and rotational degrees of freedom influence the specific heat at lower temperatures than vibrational ones, we need *two* rotational degrees of freedom, i.e. a diatomic gas, where the moments of inertia around the two perpendicular axes to the axis of symmetry of the molecule are nonzero.

Problem 1.4 First law of thermodynamics revision – isothermal

A quantity of gas at 120 bar, 0.8 m³ and 25°C is compressed isothermally to 0.2 m³. Calculate

- (i) the change in internal energy of the gas, The gas is ideal, so $\Delta T = 0 \Rightarrow \Delta U = 0$.

(ii) the work done on the gas,

$$W = - \int_{V_1}^{V_2} p dV \quad . (\text{Inserting } pV = nRT)$$

$$\Rightarrow W = -nRT \int_{V_1}^{V_2} \frac{dV}{V} = nRT \ln \frac{V_1}{V_2} .$$

From initial conditions, $nRT = p_1 V_1$, so

$$W = 120 \times 10^5 \times 0.8 \times \ln 4 = 13.3 \text{ MJ}.$$

(iii) the heat flow and which way it goes. The first law says $\Delta U = Q + W = 0$ from part a), therefore $Q = -W = -13.3 \text{ MJ}$.

Problem 1.5 First law of thermodynamics revision – adiabatic

4 m³ of gas at 50 bar and 120°C expands adiabatically to twice the initial volume. Molar heat capacities, $c_v = 20.8 \text{ J K}^{-1} \text{ mol}^{-1}$, $c_p = 29.1 \text{ J K}^{-1} \text{ mol}^{-1}$. Calculate

(i) the work done by the gas, by direct integration of the work, Using that $pV^\gamma = C$ with C a constant, integrating the work gives

$$W = \int_{V_1}^{V_2} p dV$$

$$= \int_{V_1}^{V_2} V^{-\gamma} C dV$$

$$= \frac{C V_2^{1-\gamma} - C V_1^{1-\gamma}}{1-\gamma} .$$

Now choose $C = p_2 V_2^\gamma$ in the first term and $C = p_1 V_1^\gamma$ in the second (they are equal!) then we see a simplification:

$$W = \frac{p_2 V_2^\gamma V_2^{1-\gamma} - p_1 V_1^\gamma V_1^{1-\gamma}}{1-\gamma}$$

$$= \frac{p_2 V_2 - p_1 V_1}{1-\gamma}$$

$$= \frac{nR}{1-\gamma} (T_2 - T_1)$$

$$= \frac{C_v nR}{C_v - C_p} (T_2 - T_1)$$

$$(\text{C}_v - \text{C}_p = -nR.) \Rightarrow W = -C_v (T_2 - T_1) > 0 . \quad (1.1)$$

So the work done only depends on the initial and final temperatures. This simplifies discussion of Carnot cycles as we shall see.

For the numerical evaluation, use $C = p_1 V_1^\gamma$ in both terms then: plugging in the numbers with $\gamma = 29.1/20.8 = 1.4$:

$$W = \frac{50 \times 10^5 \times 4^{1.4}}{-0.4} [8^{-0.4} - 4^{-0.4}] = 12.1 \text{ MJ}.$$

(ii) the final temperature. On an adiabatic transformation $TV^{1-\gamma}$ is constant, therefore

$$T_2 = \left(\frac{V_2}{V_1}\right)^{1-\gamma} T_1 = 2^{-0.4}(273 + 120) = 298 \text{ K} = 25^\circ\text{C}.$$

(iii) the change in energy, using part (ii). First we evaluate the amount of substance with the equation of state: $n = (p_1 V_1)/(RT_1)$. For an ideal gas, $U = C_V T$, therefore here

$$\Delta U = n c_V \Delta T = \frac{50 \times 10^5 \times 4}{8.31 \times (273 + 120)} 20.8 \times (120 - 25) = 12.1 \text{ MJ}.$$

Of course, we retrieve the result from part a) since $\Delta U = W$ in an adiabatic transformation. The method of calculation of W in part (iii) is simpler.

Problem 1.6 Frictional isothermal piston

0.1 mol of an ideal gas in a cylindrical container of unit cross-sectional area expands isothermally and quasi-statically at 20°C from 2 m^3 to 4 m^3 , doing work against a piston. There is friction between the piston and the cylinder wall that provides a force of 10 N when the piston is moving. How much external heat is needed to accomplish the expansion assuming the heat generated by the friction all goes into the gas? [148.8 J]

With or without friction, since $\Delta T = 0$ (isothermal), the internal energy is constant and from the first law, the provided heat (not counting the friction heat) must balance out the work output from the piston (the work done by the force that is reduced by the friction).

Let's first consider a transformation *without friction* that takes the gas to the *same* final state, that is to say the same volume. In this case, the (reversible) work done *by* the expanding gas is

$$\begin{aligned} W_{\text{by}} &= \int_{V_1}^{V_2} p \, dV \\ &= nRT \int_{V_1}^{V_2} \frac{dV}{V} \\ &= nRT \ln \frac{V_2}{V_1} \\ &= 0.1 \times 8.31 \times (273 + 20) \ln 2 = 168.8 \text{ J}. \end{aligned}$$

With friction, the work done *by the gas* is the same, but the gas pressure now needs to overcome the friction force, and only the remainder is available to the surroundings. This means that compared to the frictionless case, the usable work is reduced by $10 \times 2/1 = 20 \text{ J}$ to 148.8 J. The same amount of heat (148.8 J) as the *usable* work is therefore needed to perform the expansion, as the work performed against friction becomes heat and is returned to the gas. Conservation of energy is obeyed (which is the essence of the First Law): the heat input is equal to the output work.

If the pressure is reduced towards zero, at some point it is less than the maximum frictional force (10 N). At that point there is no expansion and no work delivered. There will be hysteresis, if the moving coefficient of friction is less than the stationary one. As the pressure is *increased* once moving it will be able to perform some work (as the frictional force will drop once moving); but as the pressure is lowered, the amount of work performed will tend to zero (this will occur at a lower pressure than that required to start it moving).

Problem 1.7 A constant-force frictional piston

An monoatomic ideal gas in a cylindrical container of unit cross-sectional area expands quasi-statically from 2 m^3 to 4 m^3 while pushing on a piston that applies a constant force of 100 N on external machinery. How much external heat is needed to accomplish the expansion when:

- (i) there is no friction between the piston and the cylinder wall?
- (ii) the friction between the piston and the cylinder wall provides a force of 10 N when the piston is moving? Assume that the heat generated by the friction all goes into the gas.

The answer to the previous problem may seem counter-intuitive because one may expect a piston with friction to be less effective than a piston without friction. And it is, the key is to realise that in the configuration of question 6, the usable work is smaller when there is some friction. Problem 7 answers the question of what happens when the usable work stays the same, here 100 N of constant force over the displacement due to the expansion.

- (i) The first thing to notice is that we are providing the heat at constant pressure, which, for a monoatomic gas, gives

$$Q = C_p \Delta T = \frac{5}{2} nR \Delta T.$$

The change in temperature can be obtained using the usual equation of state for an ideal gas, as $T = pV/(nR)$. Note that p is constant, but not V . So unlike for the previous question, the temperature is not constant during the expansion. It changes by $\Delta T = p \Delta V/(nR)$.

The transformation is quasi-static, so the pressure of the gas is the same as the externally applied pressure: $p = 100 \text{ N m}^{-2}$. This gives a total heat to be provided to the gas of

$$Q = \frac{5}{2} nR \Delta T = \frac{5}{2} p \Delta V = 500 \text{ J}.$$

- (ii) When friction is present, for the same usable work, it increases the pressure by 10% to $p = 110 \text{ N m}^{-2}$, resulting in both ΔU and ΔT changing by 10%, for a provided heat of 550 J .

At this point, we have to remember that the energy dissipated in friction goes back to the gas in the form of heat, so it has to be subtracted in order to get the net heat that needs to be externally provided to the gas. The friction heat is equal to the work of the friction force, which is 10 N over 2 m , that is to say 20 J . The net heat is therefore 530 J , which this time is more than the heat for the frictionless case.